

Performance Testing

Date	29 September 2026
Team ID	SWTID-2026-8574
Project Name	LegalEase: AI-Powered Legal Document Generator
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained

Step 1: Testing Overview

Field	Details
Testing Tool Used	JMeter, Locust, Postman
Type of Testing	Load Testing, Stress Testing, Response Time
Target Module / API	/generate-document, /api/templates, /export-pdf
Test Environment	Cloud - AWS EC2 + Vercel Hosted Server
Test Date	28 September 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Generate Rental Agreement - 10 users parallel	10	60	Response <2s, PDF generated
2	Generate NDA - 50 users spike test	50	120	No failure, <5s response
3	Template list API - 100 users load	100	60	Throughput > 20 req/sec
4	Concurrent PDF export - stress test	25	180	Error rate <1%

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.42 sec	Pass	Optimal
2	Response Time (Max)	< 5 seconds	3.8 sec	Pass	Within limit
3	Throughput (Req/sec)	> 15 req/sec	24.5 req/sec	Pass	High throughput
4	Error Rate	< 1%	0.2%	Pass	Negligible errors
5	CPU Utilization	< 80%	62%	Pass	Stable
6	Memory Utilization	< 80%	58%	Pass	Efficient

Step 4: Observations & Analysis

Key Findings:
System handled 100 concurrent users with avg response 1.42s. AI generation takes 2-3s but cached templates respond <500ms. Throughput exceeded target.

Bottlenecks Identified:
LLM API call is main bottleneck under spike load (50+ users). PDF generation CPU spikes briefly to 70%. No DB bottleneck observed.

Optimization Steps Taken:
Added response caching for templates, async PDF generation, connection pooling, and rate limiting for LLM API. Reduced avg response by 35%.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

[JMeter Report - Avg Response 1.42s]

Graph: Throughput vs Users - 24.5 req/sec at 100 users

[Locust Load Test - 0.2% Error Rate]

Screenshot: AWS CloudWatch CPU 62%, Memory 58%